Supplementary tables

Table S1 | Comparison of response surface models for $\mu_{\max}$ and $N_{\max}$ across light and medium dilution factors for flask and microplate data. $R^2_{\mu_{\max}}$ cal: R^2_{param} for parameters of $\mu_{\max}$ on calibration conditions, $R^2_{\mu_{\max}}$ val: R^2_{param} for parameters of $\mu_{\max}$ on all conditions, $R^2_{N_{\max}}$ cal: R^2_{param} for parameters of $N_{\max}$ on calibration conditions, $R^2_{N_{\max}}$ val: R^2_{param} for parameters of $N_{\max}$ on all conditions. For each model, the first row corresponds to flask data and the second row to microplate data.

Model	**Mathematical expression**	$R^2_{\mu_{\max}}$ cal	$R^2_{\mu_{\max}}$ val	$R^2_{N_{\max}}$ cal	$R^2_{N_{\max}}$ val
Monod double	$\beta_0 \dfrac{L_0}{L_0+\beta_1} \dfrac{C_0}{C_0+\beta_2}$	0.851	0.515	0.917	0.955
		0.885	0.701	0.875	0.800
Monod × Haldane	$\beta_0 \dfrac{L_0}{L_0+\beta_1+\frac{L_0^2}{\beta_2}} \dfrac{C_0}{C_0+\beta_3}$	0.850	0.514	0.990	0.855
		0.891	0.740	0.931	0.810
Inverse polynomial	$\dfrac{\beta_0}{1+\beta_1 L_0+\beta_2 C_0+\beta_3 L_0 C_0}$	0.894	0.303	0.814	−0.307
		0.448	0.155	0.506	−0.214
Double Hill	$\beta_0 \dfrac{L_0^{\beta_1}}{L_0^{\beta_1}+\beta_2^{\beta_1}} \dfrac{C_0^{\beta_3}}{C_0^{\beta_3}+\beta_4^{\beta_3}}$	0.855	0.351	0.931	0.911
		0.902	0.701	0.928	0.747
Exponential saturation	$\beta_0\left(1-\exp(-\beta_1 L_0)\right)\left(1-\exp(-\beta_2 C_0)\right)$	0.819	0.286	0.991	0.850
		0.777	0.742	0.896	0.809
2D Gaussian	$\beta_0 \exp\Big(-\beta_1(L_0-\beta_2)^2-\beta_3(C_0-\beta_4)^2 -\beta_5(L_0-\beta_2)(C_0-\beta_4)\Big)$	0.944	0.673	0.999	0.811
		0.930	0.731	0.898	0.837
Steele	$\beta_0 \dfrac{L_0}{\beta_1} \exp\left(1-\dfrac{L_0}{\beta_1}\right) \dfrac{C_0}{C_0+\beta_2}$	0.808	0.403	0.811	0.922
		0.779	0.731	0.927	0.812
Monod β_0 modulation	$\left(\beta_0+\dfrac{\beta_1 C_0}{C_0+\beta_2}\right) \dfrac{L_0}{L_0+\beta_3} \dfrac{C_0}{C_0+\beta_4}$	0.851	0.515	0.985	0.885
		0.902	0.692	0.875	0.800
Bilinear logistic	$\beta_0 \dfrac{1}{1+\exp[-\beta_1(L_0-\beta_2)]} \dfrac{1}{1+\exp[-\beta_3(C_0-\beta_4)]}$	0.790	0.609	0.990	0.835
		0.914	0.730	0.945	0.744

Table S2 | Response surface estimated parameters for $\mu_{\max}$ and $N_{\max}$ as functions of light L_0 and medium dilution C_0 for flask data.

Model	Parameters ($\mu_{\max}$)	Parameters ($N_{\max}$)
Monod double	$\beta_0 = 2.84 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 4.09 \times 10^{-1}$ $\beta_2 = 4.26 \times 10^{-2}$	$\beta_0 = 2.78 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 5.90 \times 10^{-2}$ $\beta_2 = 3.02$
Monod × Haldane	$\beta_0 = 2.87 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 4.00 \times 10^{-1}$ $\beta_2 = 1.00 \times 10^{2}$ $\beta_3 = 4.30 \times 10^{-2}$	$\beta_0 = 2.22 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 0.00$ $\beta_2 = 8.00$ $\beta_3 = 2.32$
Inverse polynomial	$\beta_0 = 1.45 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 1.00 \times 10^{-3}$ $\beta_2 = 9.03 \times 10^{-1}$ $\beta_3 = -1.23$	$\beta_0 = 6.11 \times 10^{7}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 4.39$ $\beta_2 = 1.00 \times 10^{-2}$ $\beta_3 = -4.37$
Double Hill	$\beta_0 = 7.06 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 6.00 \times 10^{-1}$ $\beta_2 = 5.00$ $\beta_3 = 2.23$ $\beta_4 = 7.30 \times 10^{-2}$	$\beta_0 = 1.82 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 3.00 \times 10^{-1}$ $\beta_2 = 1.00 \times 10^{-1}$ $\beta_3 = 1.27$ $\beta_4 = 8.52 \times 10^{-1}$
Exponential saturation	$\beta_0 = 2.09 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 2.54$ $\beta_2 = 1.00 \times 10^{2}$	$\beta_0 = 1.16 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 5.00 \times 10^{2}$ $\beta_2 = 7.36 \times 10^{-1}$
2D Gaussian	$\beta_0 = 4.35 \times 10^{9}\,\mathrm{h}^{-1}$ $\beta_1 = 5.76 \times 10^{-1}$ $\beta_2 = 6.05 \times 10^{2}$ $\beta_3 = 4.96 \times 10^{-1}$ $\beta_4 = 6.48 \times 10^{2}$ $\beta_5 = -1.06$	$\beta_0 = 6.38 \times 10^{7}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 1.00 \times 10^{-1}$ $\beta_2 = 1.00 \times 10^{-1}$ $\beta_3 = 2.91$ $\beta_4 = 9.20 \times 10^{-1}$ $\beta_5 = 2.29 \times 10^{-1}$
Steele	$\beta_0 = 2.02 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 1.10$ $\beta_2 = 4.30 \times 10^{-2}$	$\beta_0 = 3.32 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 5.00 \times 10^{-1}$ $\beta_2 = 3.27$
Monod β_0 modulation	$\beta_0 = 2.84 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 2.86 \times 10^{-11}$ $\beta_2 = 2.00$ $\beta_3 = 4.00 \times 10^{-1}$ $\beta_4 = 4.30 \times 10^{-2}$	$\beta_0 = 3.54 \times 10^{7}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 1.01 \times 10^{8}$ $\beta_2 = 5.91 \times 10^{-1}$ $\beta_3 = 0.00$ $\beta_4 = 5.91 \times 10^{-1}$
Bilinear logistic	$\beta_0 = 3.51 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 3.25$ $\beta_2 = 3.00 \times 10^{-1}$ $\beta_3 = 4.71 \times 10^{-1}$ $\beta_4 = 1.00 \times 10^{-2}$	$\beta_0 = 1.23 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 1.00 \times 10^{-1}$ $\beta_2 = 0.00$ $\beta_3 = 5.26$ $\beta_4 = 4.32 \times 10^{-1}$

Table S3 | Response surface estimated parameters for $\mu_{\max}$ and $N_{\max}$ as functions of light L_0 and medium dilution C_0 for microplate data.

Model	**Parameters ($\boldsymbol{\mu}_{\mathbf{max}}$)**	**Parameters ($\boldsymbol{N}_{\mathbf{max}}$)**
Monod double	$\beta_0 = 1.50 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 9.20 \times 10^{-2}$ $\beta_2 = 5.82 \times 10^{-2}$	$\beta_0 = 4.60 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 1.46 \times 10^{-1}$ $\beta_2 = 3.77$
Monod × Haldane	$\beta_0 = 1.95 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 1.69 \times 10^{-1}$ $\beta_2 = 3.70$ $\beta_3 = 5.80 \times 10^{-2}$	$\beta_0 = 1.58 \times 10^{9}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 7.43 \times 10^{-1}$ $\beta_2 = 5.00 \times 10^{-1}$ $\beta_3 = 4.45$
Inverse polynomial	$\beta_0 = 8.30 \times 10^{-2}\,\mathrm{h}^{-1}$ $\beta_1 = 1.00 \times 10^{-3}$ $\beta_2 = 1.00 \times 10^{-2}$ $\beta_3 = -4.37 \times 10^{-1}$	$\beta_0 = 5.49 \times 10^{7}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 7.89$ $\beta_2 = 1.00 \times 10^{-2}$ $\beta_3 = -8.20$
- Double Hill	$\beta_0 = 1.79 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 1.19$ $\beta_2 = 9.30 \times 10^{-2}$ $\beta_3 = 5.60 \times 10^{-1}$ $\beta_4 = 9.40 \times 10^{-2}$	$\beta_0 = 4.64 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 4.00$ $\beta_2 = 1.57 \times 10^{-1}$ $\beta_3 = 9.60 \times 10^{-1}$ $\beta_4 = 5.00$
Exponential saturation	$\beta_0 = 1.25 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 7.28$ $\beta_2 = 1.00 \times 10^{1}$	$\beta_0 = 2.31 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 5.70$ $\beta_2 = 4.43 \times 10^{-1}$
2D Gaussian	$\beta_0 = 1.43 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 1.12$ $\beta_2 = 7.49 \times 10^{-1}$ $\beta_3 = 1.08$ $\beta_4 = 7.70 \times 10^{-1}$ $\beta_5 = 1.05 \times 10^{-1}$	$\beta_0 = 8.89 \times 10^{7}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 2.11$ $\beta_2 = 6.63 \times 10^{-1}$ $\beta_3 = 4.12$ $\beta_4 = 8.59 \times 10^{-1}$ $\beta_5 = 1.16$
Steele	$\beta_0 = 1.46 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 6.13 \times 10^{-1}$ $\beta_2 = 5.80 \times 10^{-2}$	$\beta_0 = 4.97 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 5.96 \times 10^{-1}$ $\beta_2 = 4.70$
Monod β_0 modulation	$\beta_0 = 8.10 \times 10^{-2}\,\mathrm{h}^{-1}$ $\beta_1 = 8.43 \times 10^{-2}$ $\beta_2 = 2.58 \times 10^{-1}$ $\beta_3 = 9.20 \times 10^{-2}$ $\beta_4 = 1.10 \times 10^{-2}$	$\beta_0 = 4.60 \times 10^{8}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 5.46 \times 10^{-1}$ $\beta_2 = 2.00$ $\beta_3 = 1.46 \times 10^{-1}$ $\beta_4 = 3.77$
Bilinear logistic	$\beta_0 = 1.33 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1 = 6.07$ $\beta_2 = 2.90 \times 10^{-2}$ $\beta_3 = 5.66$ $\beta_4 = 1.00 \times 10^{-2}$	$\beta_0 = 9.41 \times 10^{7}\,\mathrm{cell\,mL}^{-1}$ $\beta_1 = 3.11 \times 10^{2}$ $\beta_2 = 1.56 \times 10^{-1}$ $\beta_3 = 4.01$ $\beta_4 = 5.51 \times 10^{-1}$

Table S4 | Estimated parameters for microalgae growth models in flask culture under light variation and medium dilution.

Model acronym	Parameters
LMMB	$\mu_{\max} = 3.03 \times 10^{-1}\,\mathrm{h}^{-1}$ $\xi_C = 3.14 \times 10^{-2}$ $\lambda_L = 7.50 \times 10^{-7}\,\mathrm{\mu mol_{h\nu}\,m^{-2}\,s^{-1}\,cell^{-1}\,mL}$ $\Gamma = 4.96 \times 10^{7}\,\mathrm{cell\,mL^{-1}}$ $\alpha = 1.70 \times 10^{-8}\,\mathrm{cell^{-1}\,mL}$ $K = 5.08 \times 10^{-9}\,\mathrm{mL\,cell^{-1}\,cm^{-1}}$
MMmB	$\mu_{\max} = 2.85 \times 10^{-1}\,\mathrm{h}^{-1}$ $K_C = 4.70 \times 10^{6}$ $K_I = 3.10 \times 10^{1}\,\mathrm{\mu mol_{h\nu}\,m^{-2}\,s^{-1}}$ $m = 3.60 \times 10^{-3}\,\mathrm{h}^{-1}$ $\alpha = 1.57 \times 10^{-8}\,\mathrm{cell^{-1}\,mL}$ $\beta = 1.50 \times 10^{-4}\,\mathrm{h^{-1}\,\mu mol_{h\nu}^{-1}\,m^{2}\,s}$ $K = 5.11 \times 10^{-6}\,\mathrm{mL\,cell^{-1}\,cm^{-1}}$
DMB	$\mu_{\max} = 3.61 \times 10^{-1}\,\mathrm{h}^{-1}$ $Q_L = 2.14\,\mathrm{cell^{-1}\,mL}$ $Q_0 = 3.68 \times 10^{-1}\,\mathrm{cell^{-1}\,mL}$ $K_I = 1.06 \times 10^{2}\,\mathrm{\mu mol_{h\nu}\,m^{-2}\,s^{-1}}$ $K_C = 3.45 \times 10^{7}$ $K = 1.30 \times 10^{-5}\,\mathrm{mL\,cell^{-1}\,cm^{-1}}$ $\rho_{\max} = 5.37 \times 10^{-1}\,\mathrm{h^{-1}\,cell^{-1}\,mL}$
Hybrid RSM–ODE	$\beta_0^{(\mu)} = 2.83 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1^{(\mu)} = 4.26 \times 10^{-2}$ $\beta_2^{(\mu)} = 4.09 \times 10^{-1}$ $\beta_0^{(N)} = 2.60 \times 10^{8}\,\mathrm{cell\,mL^{-1}}$ $\beta_1^{(N)} = 3.02$ $\beta_2^{(N)} = 5.88 \times 10^{-2}$

Table S5 | Estimated parameters for microalgae growth models in plate cultures under light variation and medium dilution.

Model acronym	Parameters
LMMB	$\mu_{\max} = 1.67 \times 10^{-1}\,\mathrm{h}^{-1}$ $\xi_C = 8.00 \times 10^{-3}$ $\lambda_L = 1.20 \times 10^{-7}\,\mu\mathrm{mol}_{h\nu}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}\,\mathrm{cell}^{-1}\,\mathrm{mL}$ $\Gamma = 6.40 \times 10^{7}\,\mathrm{cell}\,\mathrm{mL}^{-1}$ $\alpha = 1.30 \times 10^{-8}\,\mathrm{cell}^{-1}\,\mathrm{mL}$ $K = 1.00 \times 10^{-8}\,\mathrm{mL}\,\mathrm{cell}^{-1}\,\mathrm{cm}^{-1}$
MMmB	$\mu_{\max} = 1.55 \times 10^{-1}\,\mathrm{h}^{-1}$ $K_C = 3.00 \times 10^{6}$ $K_I = 1.40 \times 10^{1}\,\mu\mathrm{mol}_{h\nu}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}$ $m = 3.00 \times 10^{-3}\,\mathrm{h}^{-1}$ $\alpha = 1.29 \times 10^{-8}\,\mathrm{cell}^{-1}\,\mathrm{mL}$ $\beta = 0.00\,\mathrm{h}^{-1}\,\mu\mathrm{mol}_{h\nu}^{-1}\,\mathrm{m}^{2}\,\mathrm{s}$ $K = 5.98 \times 10^{-6}\,\mathrm{mL}\,\mathrm{cell}^{-1}\,\mathrm{cm}^{-1}$
DMB	$\mu_{\max} = 2.87 \times 10^{-1}\,\mathrm{h}^{-1}$ $Q_L = 6.50 \times 10^{-1}\,\mathrm{cell}^{-1}\,\mathrm{mL}$ $Q_0 = 1.32 \times 10^{-1}\,\mathrm{cell}^{-1}\,\mathrm{mL}$ $K_I = 5.80 \times 10^{1}\,\mu\mathrm{mol}_{h\nu}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}$ $K_C = 2.47 \times 10^{7}$ $K = 4.12 \times 10^{-5}\,\mathrm{mL}\,\mathrm{cell}^{-1}\,\mathrm{cm}^{-1}$ $\rho_{\max} = 1.90 \times 10^{-1}\,\mathrm{h}^{-1}\,\mathrm{cell}^{-1}\,\mathrm{mL}$
Hybrid RSM–ODE	$\beta_0^{(\mu)} = 1.95 \times 10^{-1}\,\mathrm{h}^{-1}$ $\beta_1^{(\mu)} = 1.69 \times 10^{-1}$ $\beta_2^{(\mu)} = 3.67$ $\beta_3^{(\mu)} = 5.77 \times 10^{-2}$ $\beta_0^{(N)} = 1.58 \times 10^{9}\,\mathrm{cell}\,\mathrm{mL}^{-1}$ $\beta_1^{(N)} = 7.43 \times 10^{-1}$ $\beta_2^{(N)} = 5.00 \times 10^{-1}$ $\beta_3^{(N)} = 4.45$